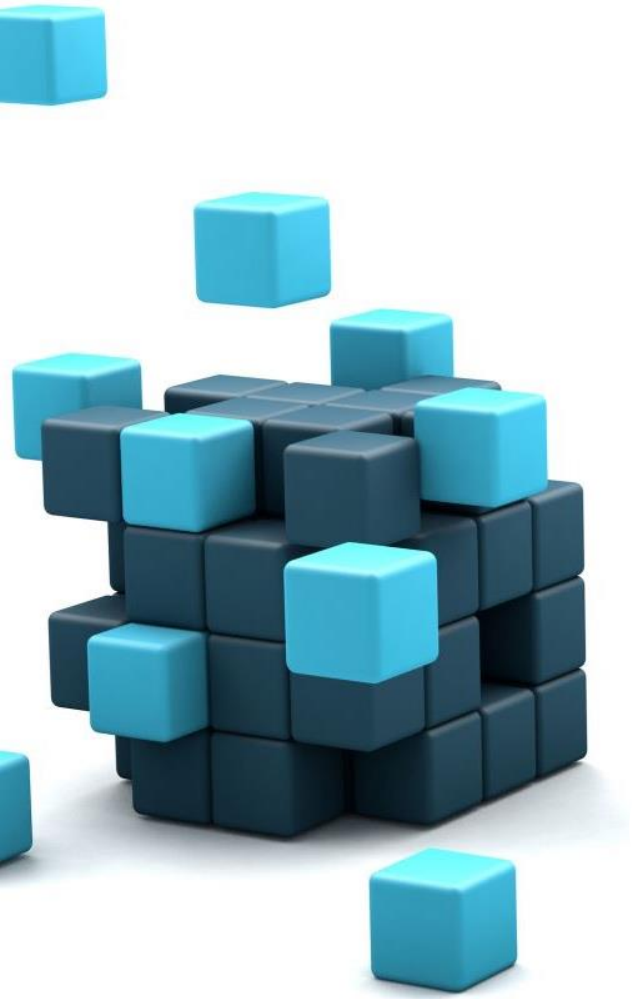
Day 2: Pandas Data Handling

- Load CSV files
- Data Frames
- Filtering & inspection



Introduction to Data Analysis in Python

What is Pandas?

- A Python library for working with structured data
- Used for data analysis and manipulation
- Works with tabular data (rows & columns)

Reading Data into Python

What is a CSV file?

- Comma-Separated Values file
- Common format for datasets

How Pandas loads data:

- Reads files into a DataFrame

Example:

- pd.read_csv("file.csv")

Output:

- Dataset becomes usable in Python

Core Structure in Pandas

What is a DataFrame?

- A table-like structure (rows & columns)
- Similar to Excel spreadsheet

Key Features:

- Columns represent variables
- Rows represent records

Example:

- Student dataset
- Sales dataset
- Loan dataset

Exploring Your Data

Inspection Methods:

- View first rows → head()
- View last rows → tail()
- Check structure → info()
- Summary stats → describe()

Filtering Data:

- Select specific columns
- Filter rows based on conditions

Example:

- Age > 25
- Salary > 50,000

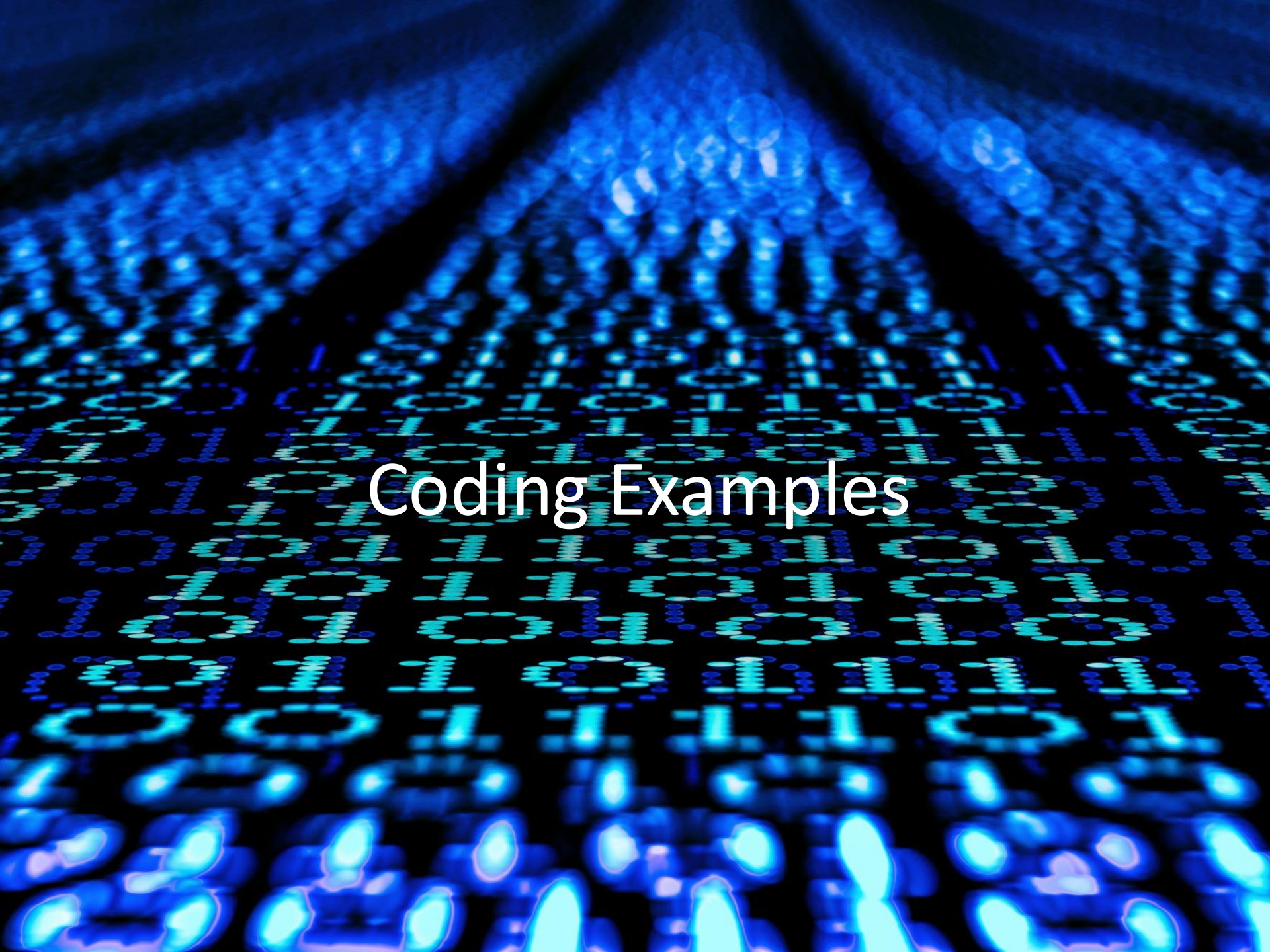
What You Learned

HOW TO LOAD DATASETS USING PANDAS

UNDERSTANDING DATAFRAMES

INSPECTING DATA FOR STRUCTURE AND QUALITY

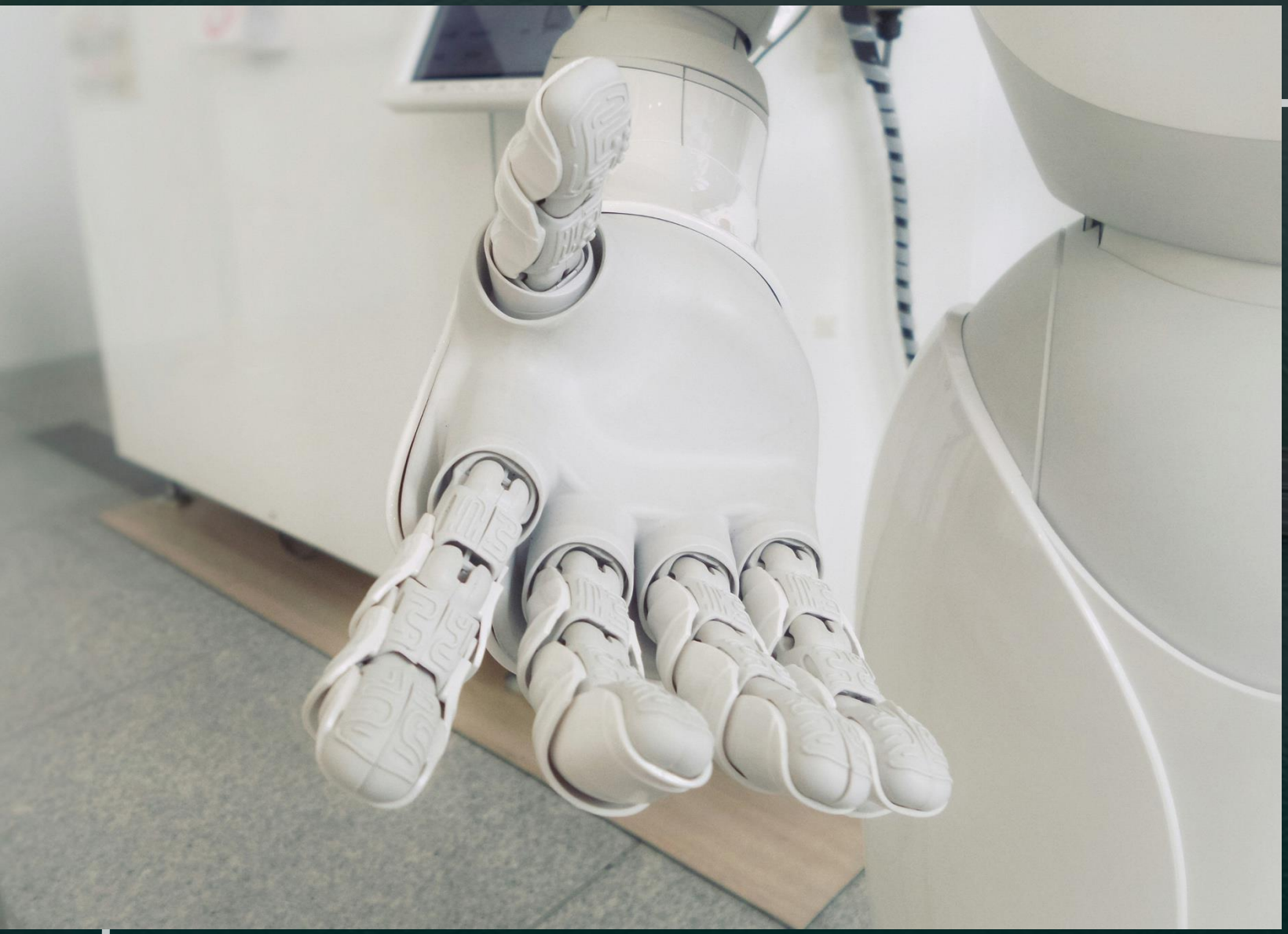
FILTERING DATA FOR ANALYSIS



Coding Examples

Review + Resources

- https://docs.python.org/3/tutorial/
- https://numpy.org/doc/stable/user/quickstart.html
- https://www.w3schools.com/python/numpy/

GitHub materials: https://github.com/AIWAZIRI-LABS/Introduction-to-AI-Machine-Learning-2026-ITng

Youtube: Numpy vs pandas: https://youtu.be/KHoEbRH46Zk?si=VAmS1R86IzObQ5Nf